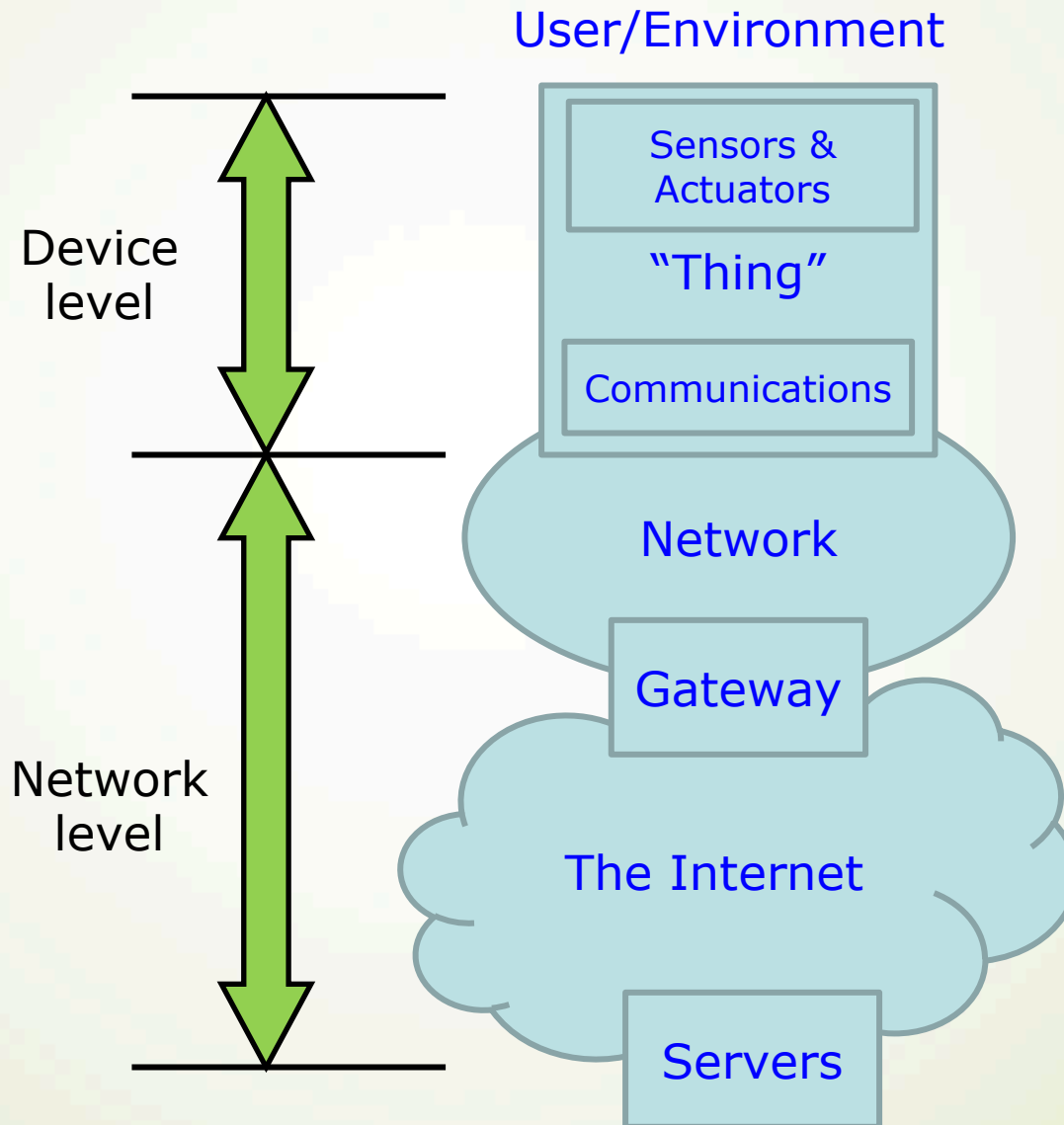


Agenda

- Internet-of-Things (IoT)
- IoT Versus Cyber-Physical Systems (CPS)
- IoT Data Processing
- IoT Analytics

IoT Hardware Platforms



Sensors and Actuators

- Sensors: input components
 - Passive, omnidirectional (e.g., mic)
 - Passive, narrow-beam sensor (e.g., PIR)
 - Active sensors (e.g., sonar, lidar, and radar)
- Actuators: output components
 - Adding lighting, heat, sound, etc.
 - Controlling motors to move objects
 - Displaying messages
- A “thing” generally consists of four main parts:
 - Sensors & actuators
 - Microcontroller
 - Network interfaces
 - Power supply

(Unique) IoT Software

- Operating systems: Contiki, TinyOS, and Linux variants
- Network protocols: 6LoWPan, CoAP
- Gateway/middleware
- Different from regular networks
 - In conventional networks, the target is **moving bits** from one place to another
 - IoTs are expected to facilitate providing **meaningful information/actions**
- Data Processing
 - Sensing and data collection, sensor data, and data-centric networks
 - Access, subscription, and integration
 - Data processing and stream data analysis
 - Query and discovery
 - Data classification and clustering

Challenges with IoT Data

- Multi-modal and heterogeneous
- Noisy and incomplete
- Time and location dependent
- Dynamic and varies in quality
- Crowded sourced data can be unreliable
- Requires (near-) real-time analysis
- Privacy and security are important issues

Cyber-Physical Systems (CPS)

- CPS is a system of collaborating computational elements controlling physical entities
- CPS integrates computation, networking, and physical processes more closely
- So what's new?
 - Highly pervasive
 - Highly automated
 - More decentralized in networking and computation

CPS Example: Autonomous Cars



CPS versus IoT

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- Appearance:
 - Internet of Things: in **1999**, introduced by Kevin Ashton
 - Cyber-Physical Systems: around **2006**, coined by Helen Gill at the National Science Foundation
- They are closely related, sometimes difficult to distinct. Here are what people say:



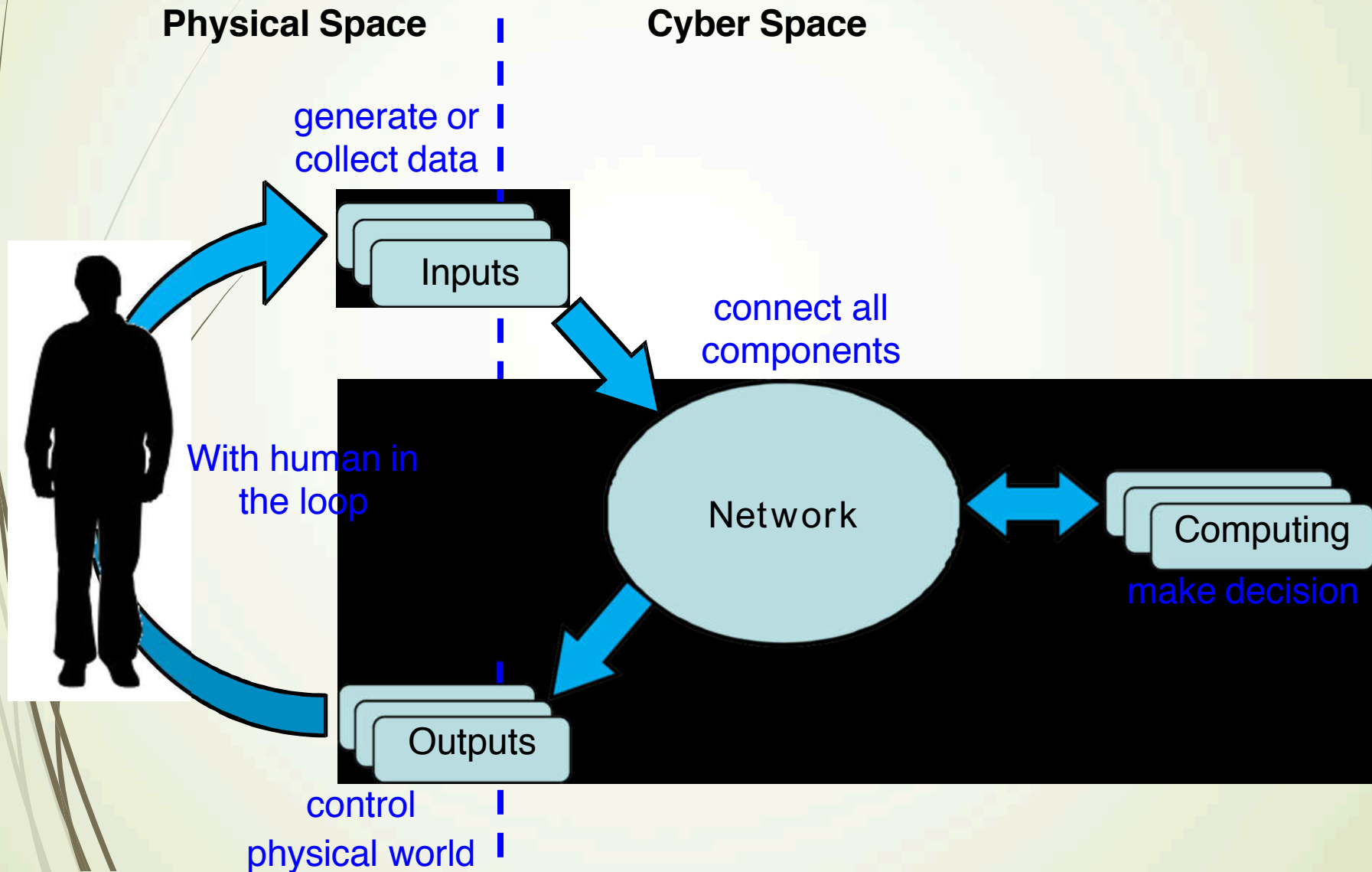
IoT has a wider scope

"CPS is the US version of IoT"



Source: Acatech 2011

CPS Model



CPS is Multidisciplinary

